Exercises

Problem 4.1: Vanishing mean score

Let $p_\varphi(x)$ be a familiy of probability distributions over x parametrised by some real parameter φ. Define the *score* as $S_\varphi(x) = \frac{\partial}{\partial\varphi}\log(p_\varphi(x))$. Show that the mean value of the score over the distribution $p_\varphi(x)$ is zero.

Problem 4.2: Additivity of Fisher information

Prove that the Fisher information is additive for independent measurements. That is, when φ is inferred from two measurements X and Y with $p_\varphi(x,y) = p_\varphi(x)p_\varphi(y)$ show that $\mathcal{F}_\varphi(X,Y) = \mathcal{F}_\varphi(X) + \mathcal{F}_\varphi(Y)$.

Problem 4.3: MSE, variance, and bias

Prove the relation (4.12) between the MSE, variance, and bias of estimators.

Problem 4.4: NOON-state interferometry

Consider an N-photon NOON-state in two modes A and B.

$$|\psi_{NOON}\rangle_{\mathrm{AB}} = \frac{1}{\sqrt{2}}(|N,0\rangle_{\mathrm{AB}} + |0,N\rangle_{\mathrm{AB}}).$$

A phase shift φ is applied to mode B and the two modes are subsequently interfered on a balanced beam splitter. The unitary encoding the phase is $\hat{U}_\varphi = e^{-i\varphi\hat{n}_{\mathrm{B}}}$, where $\hat{n}_{\mathrm{B}}$ is the number operator on mode B. At each output of the beam splitter, the photon number is measured.

(a) Show that the state after the phase shift and before the beam splitter is

$$\frac{1}{\sqrt{2}}(|N,0\rangle_{\mathrm{AB}} + e^{-i\varphi N}|0,N\rangle_{\mathrm{AB}}).$$

(b) Show that in one of the output detectors, the probabilities for observing even and odd numbers of photons are $p_{\mathrm{even}} = \cos^2(\varphi N/2)$ and $p_{\mathrm{odd}} = \sin^2(\varphi N/2)$, respectively.

(c) Show that the Fisher information corresponding to this binary measurement of photon number parity equals $\mathcal{F}_\varphi = N^2$.

Problem 4.5: QFI for pure states

Consider a noiseless quantum parameter estimation scheme where the parameter is encoded unitarily. For a pure input state $|\psi\rangle$, the output state is

$$|\psi_\varphi\rangle = \hat{U}_\varphi|\psi\rangle = e^{-i\varphi\hat{H}}|\psi\rangle,$$

where $\hat{H}$ is some hermitian operator.

(a) Show that the QFI can be expressed in terms of the variance of $\hat{H}$ in the input state $|\psi\rangle$ as

$$\mathcal{Q}_\varphi = 4\left[\langle\psi|\hat{H}^2|\psi\rangle - \langle\psi|\hat{H}|\psi\rangle^2\right] = 4\langle\Delta^2\hat{H}\rangle_\psi.$$

(b) Assume that the system is finite dimensional and that the smallest and largest eigenvalues of $\hat{H}$ are $\lambda_{\min}$ and $\lambda_{\max}$, respectively. Show that the

Exercises

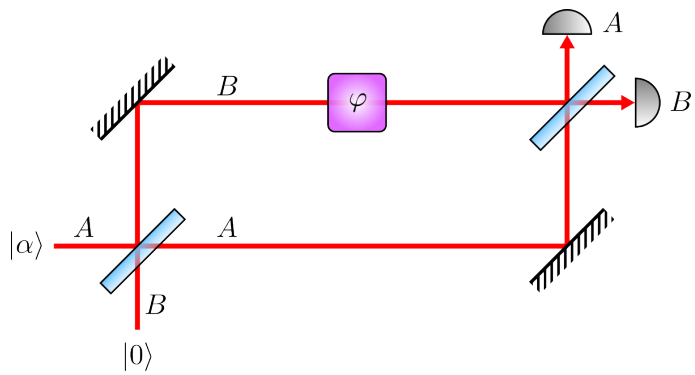

Figure 5.1: Mach-Zender interferometer with a coherent state entering one input port. Both beam splitters are balanced.

Problem 5.1: Coherent and squeezed-state interferometry

Consider a Mach-Zender interferometer as shown in Fig. 5.1, where a coherent state $|\alpha\rangle$ enters one input while vacuum is injected in the other input port. A small phase shift φ is applied in one of the interferometer arms. We take the beam splitters to be described by the unitary $\hat{U}_{BS}(\frac{1}{2})$ in Eqs. (3.134)-(3.135) (with the roles of A and B reversed at the second beam splitter) and the unitary encoding the phase shift to be $\hat{R}(-\varphi)$ (c.f. Sec. 3.1.4).

(a) Show that the probability of detecting n photons at output A as a function of φ, assuming an ideal photon counter and no losses, is given by

$$p_\varphi(n) = e^{-|\alpha|^2 \sin^2(\frac{\varphi}{2})} \frac{|\alpha|^{2n} \sin^{2n}(\frac{\varphi}{2})}{n!}. \tag{5.1}$$

(b) Find the Fisher information corresponding to the photon counting measurement. Show that the estimate variance for φ obeys the shot noise limit.

$$\langle \Delta^2 \tilde{\varphi} \rangle \geq \frac{1}{|\alpha|^2}. \tag{5.2}$$

(c) Consider now that the vacuum input in B is replaced by some arbitrary state $|\psi\rangle$. Assuming that α is real, show that the expected number of photons at output A can be written

$$\langle \hat{n}_A \rangle = \alpha^2 \sin^2(\frac{\varphi}{2}) + \langle \hat{n}_B \rangle_\psi \cos^2(\frac{\varphi}{2}) - \sqrt{2}\alpha \sin\Big(\frac{\varphi}{2}\Big) \cos\Big(\frac{\varphi}{2}\Big) \langle \hat{p}_B \rangle_\psi, \tag{5.3}$$

where $\langle \cdot \rangle_\psi = \langle \psi | \cdot | \psi \rangle$ denotes the expectation value in state $|\psi\rangle$ and $\hat{n}_B$ and $\hat{p}_B$ are the number and phase quadrature operators of mode B, respectively.

(d) Assume that $|\psi\rangle$ has support only on even photon-number (Fock) states, as is the case e.g. for vacuum and squeezed-vacuum states. Show that then

$$\begin{aligned}\langle \hat{n}_A^2\rangle =& \sin^4(\frac{\varphi}{2})(\alpha^2+\alpha^4)+\cos^4(\frac{\varphi}{2})\langle \hat{n}_B^2\rangle_\psi \\ &+2\sin^2(\frac{\varphi}{2})\cos^2(\frac{\varphi}{2})\alpha^2\langle \hat{n}_B\rangle_\psi \\ &+\sin^2(\frac{\varphi}{2})\cos^2(\frac{\varphi}{2})\left[2\alpha^2\langle \hat{p}_B^2\rangle_\psi+\langle \hat{n}_B\rangle_\psi\right].\end{aligned} \tag{5.4}$$

(e) Show that for vacuum input, i.e. $|\psi\rangle = |0\rangle$, we have the mean and variance at output A

$$\langle \hat{n}_A\rangle = \alpha^2\sin^2(\frac{\varphi}{2}), \tag{5.5}$$

$$\langle \Delta^2\hat{n}_A\rangle = \alpha^2\sin^2(\frac{\varphi}{2}). \tag{5.6}$$

(f) Let now $|\psi\rangle$ be a phase-squeezed vacuum state (i.e. squeezed in $\hat{p}$) with squeezing parameter r. Show that for sufficiently small phase φ and large α we have

$$\langle \hat{n}_A\rangle \approx \frac{1}{4}\alpha^2\varphi^2, \tag{5.7}$$

$$\langle \Delta^2\hat{n}_A\rangle \approx \frac{1}{2}\alpha^2\langle \hat{p}_B^2\rangle_\psi\varphi^2. \tag{5.8}$$

Hence, estimating φ from the photocurrent at output A gives

$$\langle \Delta^2\tilde{\varphi}\rangle = \frac{\langle \Delta^2\hat{n}_{\mathrm{A_{out}}}\rangle}{\left|\frac{\partial\langle \hat{n}_A\rangle}{\partial\varphi}\right|^2} \approx \frac{2\langle \hat{p}_B^2\rangle_\psi}{\alpha^2} = \frac{e^{-2r}}{\alpha^2}. \tag{5.9}$$

That is, the estimate variance is suppressed below shot noise by the squeezing factor.

Problem 5.2: QFI for NOON states

Show that the quantum Fisher information of the phase-dependent NOON state $\frac{1}{\sqrt{2}}(|N,0\rangle + e^{i\varphi N}|0,N\rangle)$ with respect to φ equals $\mathcal{Q}_\varphi = N^2$.

Problem 5.3: Photon loss in interferometry

We have seen that, in the absence of imperfections, interferometric phase estimation with N-photon NOON-states

$$|\psi_{NOON}\rangle = \frac{1}{\sqrt{2}}(|N,0\rangle + |0,N\rangle) \tag{5.10}$$

yields an estimate variance, for ν repetitions, scaling as

$$\langle \Delta^2\tilde{\varphi}\rangle_{NOON} \geq \frac{1}{\nu N^2}. \tag{5.11}$$

Using coherent states with mean photon number $N = |\alpha|^2$ (in a Mach-Zender

interferometer), on the other hand, yields

$$\langle \Delta^2 \tilde{\varphi} \rangle_\alpha \geq \frac{1}{\nu N}. \tag{5.12}$$

Ideally, NOON states thus achieve a quadratic improvement with the (mean) photon number. The sensitivities to loss, however, are very different for the two approaches.

(a) Find the state resulting from applying symmetric loss to the two modes of the NOON state (5.10) Argue that the loss of any non-zero number of photons will render the state useless for sensing the phase. *Hint: loss can be modelled by each mode passing through a beam splitter with transmittivity η – the other input of the beam splitter is vacuum and the other output is traced out.*

(b) Show that the estimate variance for the NOON state is bounded by

$$\langle \Delta^2 \tilde{\varphi} \rangle_{NOON} \geq \frac{1}{\nu \eta^N N^2}, \tag{5.13}$$

for efficiency η.

(c) Similarly, find the scaling of $\langle \Delta^2 \tilde{\varphi} \rangle_\alpha$ with the efficiency η, assuming symmetric losses in the two interferometer arms.

(d) When is it advantageous to use one or the other approach?